photograph, and places where it's low are darker. Gabor and many others realized, though, that intensity is only part of the information that light waves carry. We saw this, for example, in Figure 4.2b: while the interference pattern is affected by the intensity (the amplitude) of the light (higher-amplitude waves yield an overall brighter pattern), the pattern itself arises because the overlapping waves emerging from each of the slits reach their peak, their trough, and various intermediate wave heights at different locations along the detector screen. The latter information is called *phase information:* two light waves at a given point are said to be *in phase* if they reinforce each other (they each reach a peak or trough at the same time), *out of phase* if they cancel each other (one reaches a peak while the other reaches a trough), and, more generally, they have phase relations intermediate between these two extremes at points where they partially reinforce or partially cancel. An interference pattern thus records phase information of the interfering light waves.

Gabor developed a means for recording, on specially designed film, both the intensity and the phase information of light that bounces off an object. Translated into modern language, his approach is closely akin to the experimental setup of Figure 7.1, except that one of the two laser beams is made to bounce off the object of interest on its way to the detector screen. If the screen is outfitted with film containing appropriate photographic emulsion, it will record an interference pattern—in the form of minute, etched lines on the film's surface—between the unfettered beam and the one that has reflected off the object. The interference pattern will encode both the intensity of the reflected light and phase relations between the two light beams. The ramifications of Gabor's insight for science have been substantial, allowing for vast improvements in a wide range of measurement techniques. But for the public at large, the most prominent impact has been the artistic and commercial development of holograms.

Ordinary photographs look flat because they record only light intensity. To get depth, you need phase information. The reason is that as a light wave travels, it cycles from peak to trough to peak again, and so phase information—or, more precisely, phase differences between light beams that reflect off nearby parts of an object—encodes differences in how far the light rays have traveled. For example, if you look at a cat straight on, its eyes are a little farther away than its nose and this depth difference is encoded in the phase difference between the light beams' reflecting off each facial element. By shining a laser through a hologram, we are able to exploit the phase information the hologram records, and thereby add depth to the image. We've all seen the results: stunning three-dimensional projections generated from two-dimensional pieces of plastic. Note, though, that your eyes do not use phase information to see depth. Instead, your eyes use parallax: the slight difference in the angles at which light from a given point travels to reach your left eye and your right eye supplies information that your brain decodes into the point's distance. That's why, for example, if you lose sight in one eye (or just keep it closed for a while), your depth perception is compromised.